

Explore your course 2 scenario: TheLook eCommerce

Learn about the course 2 lab scenarios

This reading introduces the labs scenario for course 2, which focuses on how data is structured and organized in the cloud. As you complete these labs, you'll put your cloud data analytics knowledge and skills into practice in your role as a cloud data analyst at TheLook eCommerce. You'll gain experience with data lakehouse architecture and a variety of cloud tools that help store, analyze, and process large datasets.

Note: Be sure to review the **Lab Technical Tips** reading before you begin working on the lab.

Your role



As a cloud data analyst at TheLook eCommerce, you'll play a pivotal role in helping the merchandise team gain insights into their customer data to make data-driven decisions. While TheLook eCommerce's data team has already put a plan in place to collect and analyze relevant data, the organization continues to grow rapidly, and you need to make optimizations to meet the expanding changing needs of the organization. As part of this optimization initiative, you'll be responsible for completing tasks that help improve the way TheLook eCommerce performs data analysis using cloud tools.

Your tasks

In the following course 2 labs, you'll be working with the TheLook eCommerce team to:

- **Explore flat and nested data types in BigQuery:** In this module 1 lab, you'll troubleshoot a query for the head merchandiser to learn how data types work. You'll

also explore flat and nested data types and determine product sales in specific countries.

- **Identify different batch and streaming data sources:** In this module 1 lab, you'll run queries to help the head merchandiser examine properties for different datasets and determine if the data is collected as batch or streaming data.
- **Explore a lakehouse:** In this module 2 lab, you'll combine data from a data lake and data warehouse into a table to address some of TheLook's business concerns. You'll also use data from a CSV file to better understand when to use external and standard tables to store data in the BigQuery environment.
- **Navigate Dataplex:** In this module 3 lab, you'll use Dataplex to efficiently search for tables in BigQuery that contain information about t-shirt sales.
- **Compare data analytics with BigQuery and Dataproc:** In this module 4 lab, you'll help track the success of a pilot program by creating a report and comparing data analytics using BigQuery and a Jupyter notebook in the Dataproc environment. You'll gain experience using new cloud platforms to collect and process data.
- **Manage a partitioned table in BigQuery:** In this module 4 lab, you'll help the merchandise team better understand customer behavior by making a dashboard more efficient and cost effective. You'll analyze partitioned tables to test scanning less data for queries.

As you navigate through completing these tasks, you'll continue to build upon your knowledge on how to structure and organize cloud data. Completing these course labs will help you identify and trace data sources, write effective queries, and utilize partitioning in a cloud data analytics environment.

Note: *The story, all names, characters, and incidents portrayed in this project are fictitious. No identification with actual persons (living or deceased) is intended or should be inferred. And, the data shared in this project has been created for pedagogical purposes.*

Your deliverables

With the course 2 labs, you'll gain valuable practice and apply your new skills as you complete the following:

- Review and troubleshoot queries.
- Create and join tables using BigQuery.
- Apply table partitioning to improve efficiency of data queries.
- Utilize cloud tools to collect, process, and compare data from different sources.

Good luck! Your TheLook eCommerce colleagues are eager to leverage your expertise in analyzing and reporting data in a cloud environment.

Key takeaways

The course 2 labs incorporate valuable skills you'll use on the job as a data cloud analyst. Using cloud tools to troubleshoot queries and join and partition tables is an important part of retrieving insightful information from data. You can also add these skills to your portfolio to demonstrate your experience to potential employers.

Resources for more information

Use the **Guide to BigQuery reading** in course 2 module 1 to help support you as you work through the labs.